

DAY 2

ROUTERS AND SWITCHES

**Topic: Basic Functions, Differences,
Ports and LEDs of Routers and
Switches**

Course: Diploma in Cyber Security

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Basic Functions of Routers and Switches

Router

A router is a networking device which connects different networks. In a home network, the router normally connects the local devices to the internet. It uses IP addresses to decide where network packets should be sent.

A router can also provide Wi-Fi, depending on the model.

Example: A home Wi-Fi router connects mobile phones and computers to the internet.

Switch

A switch is mainly used to connect multiple wired devices in the same local network. It receives data and forwards it to the required device using MAC addresses.

Example: In a computer lab, a switch can connect many computers together using Ethernet cables.

Main Difference

The simple way to remember it is:

Router = connects networks

Switch = connects devices in a network

Task 1: Labelled Diagram of Router and Switch

A router and a network switch have different ports which are used for power, network connections and device management.

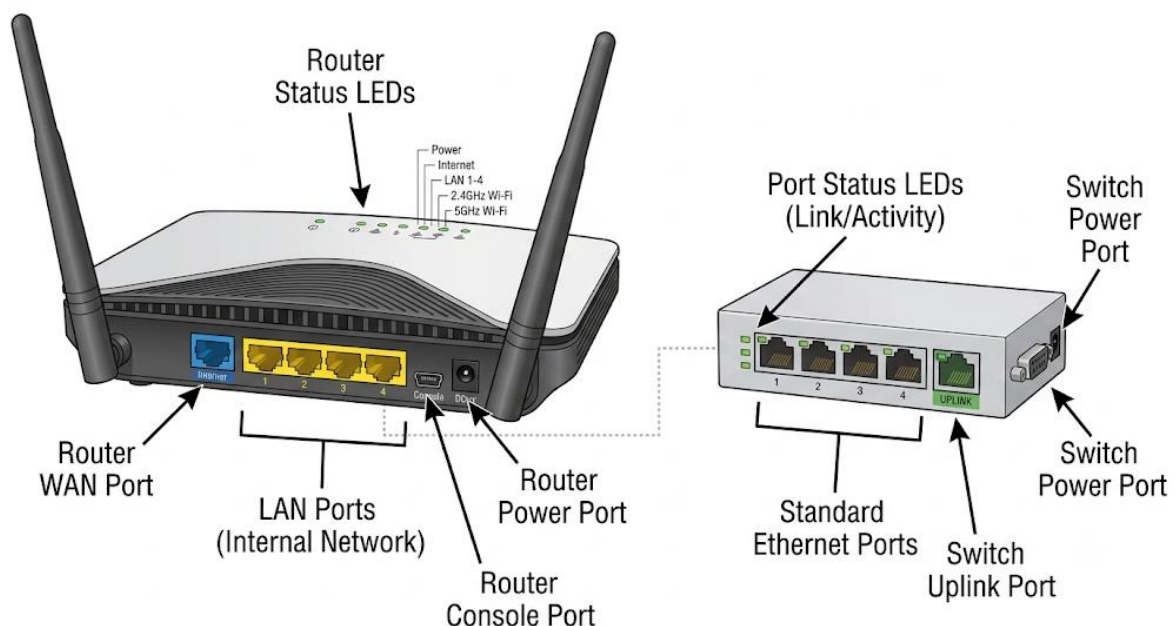
Router – Important Parts

- **WAN Port** – Used to connect the router to the modem or ISP equipment.
- **LAN Ports** – Used to connect computers and other wired devices.
- **Power Port** – Used to provide power to the router.

- **Status LEDs** – Show power, internet, Wi-Fi and LAN activity.
- **Console Port** – Available on some routers for configuration and management.

Switch – Important Parts

- **Ethernet Ports** – Used to connect computers and other network devices.
- **Uplink Port** – Used to connect the switch to another switch or router.
- **Console Port** – Available on some managed switches for configuration.
- **Power Port** – Provides power to the switch.
- **Status LEDs** – Show power and network/link activity.



Task 2: Router vs Switch

Difference	Router	Switch
Main Function	Connects different networks	Connects multiple devices in the same network
Address Used	Mainly uses IP addresses	Mainly uses MAC addresses
Traffic Handling	Routes packets between different networks	Forwards frames to the correct device within a LAN
Network Type	Used to connect different networks, such as LAN and Internet	Mainly used inside a LAN

In Simple Words

A router decides **which network the data should go to**, while a switch decides **which connected device should receive the data**.

So, a router is mainly used for communication between networks, and a switch is mainly used for communication between devices in the same local network.

Task 3: Router Identification

For this task, I used a router image and identified its important ports and LED indicators.

Parts Identified

1. Power Port

The power adapter is connected here to provide electricity to the router.

2. Ethernet Ports

The LAN Ethernet ports are used to connect wired devices such as computers or switches.

3. WAN Port

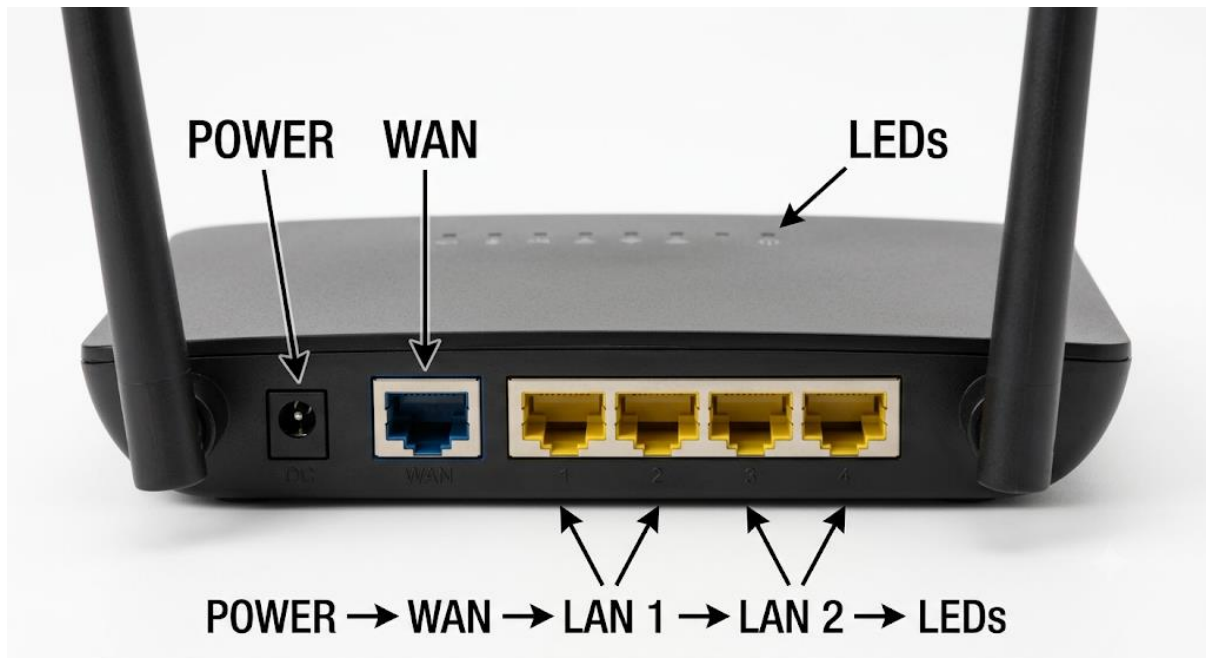
The WAN port is used for the internet-side connection, depending on the router setup.

4. LED Indicators

The LEDs show information such as power status, internet connection, Wi-Fi activity and Ethernet connection/activity.

Position

The Ethernet ports are generally located together on the back side of the router. The status LEDs are usually present on the front or top side so they can be seen easily.



Network Switch Identification

I also identified the important parts of a network switch.

Parts Identified

1. Power Port

The power cable is connected to this port to turn on the switch.

2. Ethernet Ports

These ports are used to connect computers, printers, access points and other network devices.

3. Uplink Port

An uplink connection can be used to connect the switch to a router or another switch.

4. LED Indicators

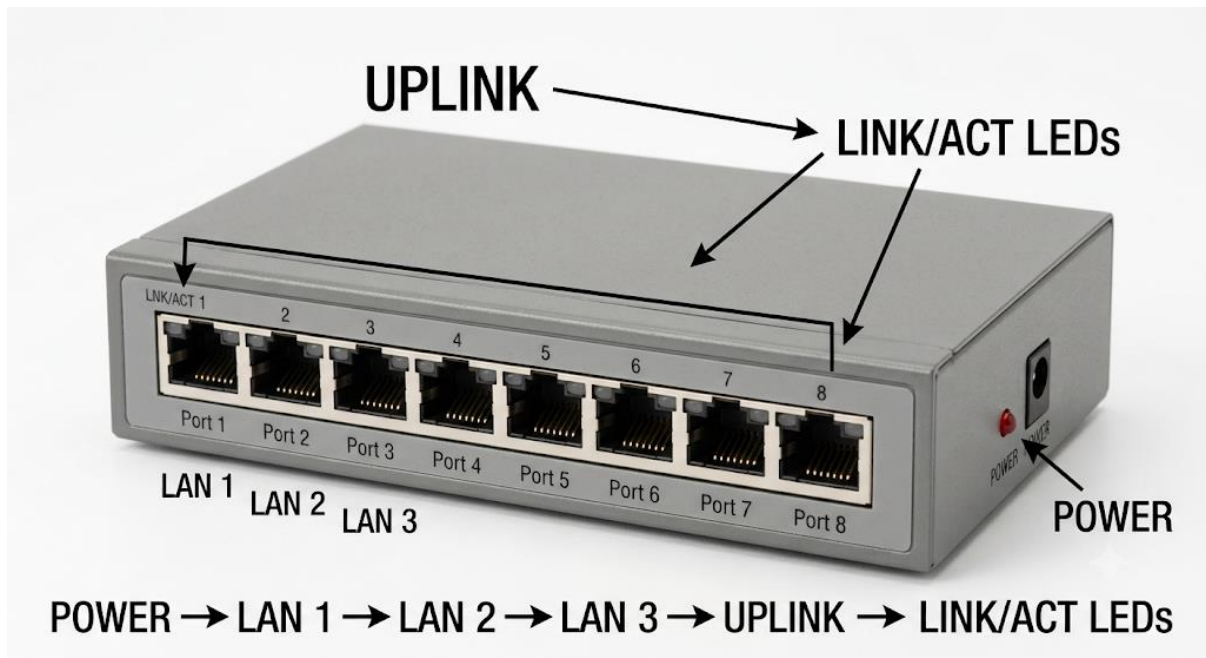
The LEDs normally show whether a port has an active connection and whether data is being transferred.

Position

The Ethernet ports are usually arranged in a row on the front panel of the switch. Small LEDs are normally located near or above the ports. The power connection is generally located on the rear side.

Practical Observation

While identifying the router and switch, I checked the ports and LEDs carefully. The LEDs are useful because they give a quick indication of power, link and network activity.



Task 4: Small Gaming Café Network

Imagine I am setting up a small gaming café where 8 gaming PCs will be used for multiplayer games and IPL streaming.

First, I would connect the ISP line to the **router**. The router will handle the internet connection and provide the local network for the café.

After that, I would connect one Ethernet cable from the router's LAN port to a **network switch**. Since I have 8 gaming PCs, I would use a switch with at least 8 available Ethernet ports. Each gaming PC would then be connected to the switch using a separate Ethernet cable.

Connection Setup

ISP / Internet



Router



Network Switch



PC 1 – PC 2 – PC 3 – PC 4 – PC 5 – PC 6 – PC 7 – PC 8

Why I would use a router first?

I would connect the router first because it acts as the connection point between the ISP/internet and the café's local network. It can manage IP addressing and route traffic between the local network and the internet.

Why use a switch for the 8 PCs?

A switch is better for connecting the 8 gaming PCs because it provides multiple wired Ethernet connections in the same LAN. Wired connections are also useful for gaming because they can provide a more stable connection compared with relying only on Wi-Fi.

Therefore, the basic setup would be:

ISP → Router → Switch → 8 Gaming PCs

This setup would allow the PCs to access the internet and communicate through the local network for multiplayer gaming.

Conclusion

From this practical, I learned the basic functions of routers and switches and how they are used in a network. I also learned how to identify important ports like WAN, LAN, Power and Uplink ports, along with the status LEDs.

I understood that a router is mainly used to connect different networks, while a switch is used to connect multiple devices within the same network. I also learned that routers mainly work with IP addresses and switches use MAC addresses for forwarding data.

The gaming café example helped me understand how a router and switch can be used together to connect multiple computers to the internet. Overall, this practical gave me a better understanding of basic networking hardware and its real-life use.